Deep Learning for Economics and Finance

Key takeaways, method choice, and outlook

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Closing lecture: synthesis, method choice, and outlook

Materials: https://github.com/sischei/Deep_Learning_for_Solving_And_Estimating_Dynamic_Economic_Models

Wrap-Up Roadmap

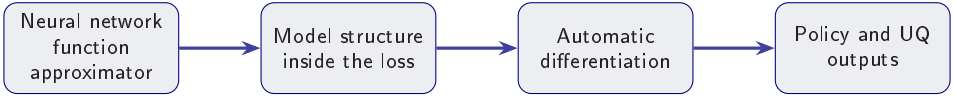
1. **Synthesis of the full course pipeline**
2. **Decision guide: DEQN, PINN, Surrogates, GP**
3. **Climate-policy lessons**
4. **Implementation checklist and outlook**

Learning objectives

By the end of the course you should be able to:

- Choose between DEQN, PINN, deep surrogates, and GP+BAL based on problem structure and scalability
- Design residual-loss functions that embed economic and climate equations for end-to-end equilibrium learning
- Articulate when to include parameters as pseudo-states versus solving repeatedly
- Evaluate whether a trained surrogate meets publication standards through residual diagnostics and stress tests
- Synthesize deep learning, uncertainty quantification, and constrained optimization for evidence-based policy design

The Big Picture: One Computational Paradigm



Across methods

DEQN, PINN, deep surrogates, and GP-BAL differ in objective and geometry, but share the same core optimization logic.

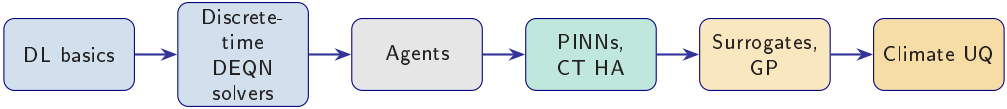
Course Arc

Day	Core step in the pipeline
1	Deep learning foundations and optimization diagnostics
2–4	DEQNs for discrete-time equilibrium (RA, IRBC)
5	Heterogeneous agents, OLG, and Young's method (sequence space, Krusell–Smith)
6	PINNs for continuous-time PDE problems
7	Deep surrogates and GP/BAL for fast policy/estimation loops
8	Climate IAMs, deep UQ, constrained policy design

Main message

The toolkit is modular: solve, emulate, quantify uncertainty, then optimize policy under explicit constraints.

Visual Timeline of the Course Workflow



Core paradigm:
model structure in loss + autodiff + scalable function approximation

Decision Guide: Which Method When?

Method	Use case	Strength
DEQN	Discrete-time equilibrium systems	Scales to high-dimensional states
PINN	Continuous-time PDE/HJB/KFE systems	Mesh-free PDE residual training
Deep surrogate	Repeated model evaluations	Very fast policy and estimation loops
GP + BAL	UQ and active design	Built-in uncertainty and sample efficiency

Brock–Mirman Through Every Lens

Method	What is approximated	Loss / objective	Diagnostic
DEQN	Savings share $s(K, z) \in (0, 1)$	Squared rel. Euler residual on simulated trajectory	$s \to \alpha\beta$ at $\delta = 1$, log utility
PINN	Value function $\hat{V}(a)$, 1-D interval	HJB residual $\rho V - \max_c[u(c) + V'(a)(ra - c)]$	$\hat{V}$ matches CRRA closed form
GP + BAL	$\hat{V}(K)$ on BM ergodic set	GP marginal likelihood; BAL picks next K_*	95% pred. band covers true V
SMM + surrogate	Policy $s(K, z, \varrho)$ pseudo-state on ϱ	SMM moment objective in ϱ	$\hat{\varrho}$ within MCSE of truth

One model, four lenses. The economic content is identical; the methodology determines what is approximated and what diagnostic certifies convergence. Companion notebooks in L03, L11, L14, L15.

When *Not* to Use Deep Learning

- **State space is genuinely low-dimensional.** $d \lesssim 5$ smooth-policy models: classical projection / VFI / perturbation usually win on accuracy and audit cost.
- **Local linearity, local question.** First/second-order responses around steady state: perturbation gives analytical IRFs in seconds (e.g. *Dynare*).
- **Likelihood-based inference + small model.** Closed-form or quadrature likelihoods: full-information ML / Bayesian particle filtering dominate any simulation-based route.
- **Small n + uncertainty is the answer.** $n \lesssim 10^3$ design points: GPs with active learning beat deep nets of comparable complexity.
- **Reproducibility / audit-ability is paramount.** Heterogeneous BLAS / GPU non-determinism makes deep-learning solutions hard to pin to the bit; deterministic FD has fewer moving parts.
- **Training instability is a model-misspecification signal.** A divergent DEQN run is usually a bad loss / missing constraint, not a missing layer.

Open Frontiers

- **Convergence theory.** Rates as a function of width, depth, and training budget.
- **Hybrid global/local.** GPs near kinks/boundary layers + DNNs for global approximation.
- **Real-time policy analysis.** Substantial speedups of full DSGE counterfactuals; carbon-tax surrogates (Kübler–Scheidegger–Surbek 2026 forthcoming, *JPE: Macroeconomics*).
- **Estimation of rich structural models.** DEQN/PINN + surrogate-based estimation pipelines (Friedl et al. 2023; Chen–Didisheim–Scheidegger 2026; Kase et al. 2022).
- **Climate–economy IAMs.** Tipping points, regional heterogeneity, multiple risks (Fernández-Villaverde–Gillingham–Scheidegger 2025).
- **Heterogeneous-agent models with aggregate shocks.** DeepHAM (Han–Yang–E 2024); DeepSAM (Payne–Rebei–Yang 2025); structural-RL price-only state (Yang–Wang–Schaab–Moll 2025).
- **Operator learning.** DeepONet (Lu et al. 2021), Fourier Neural Operator (Li et al. 2021): learn the solver, amortise across model instances.

Climate UQ in One Slide

Climate UQ in context. Pseudo-states let one trained DEQN answer policy questions for *all* plausible parameter values simultaneously. A GP surrogate on top enables fast computation of SCC, temperature paths, and welfare across the parameter space. Bayesian active learning then identifies which parameters drive policy recommendations: in the calibrations of Friedl et al. (2023), climate sensitivity is among the most important parameters for carbon-tax design (more so than the pure rate of time preference).

The payoff

Constrained Pareto-improving tax rules become feasible once the expensive solves are amortized.

Common Failure Modes and Mitigations

Deployment-time failure modes

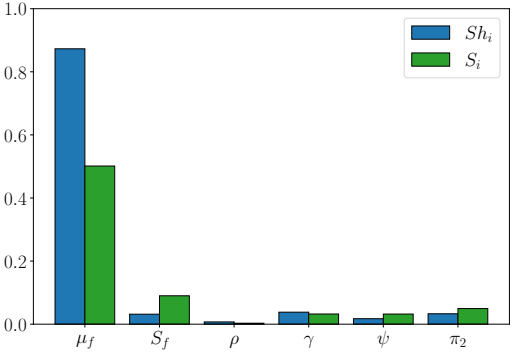
- Calibration mismatch to benchmark science
- Training focused on wrong state regions
- Over-reliance on mean loss diagnostics
- Surrogate extrapolation outside training support

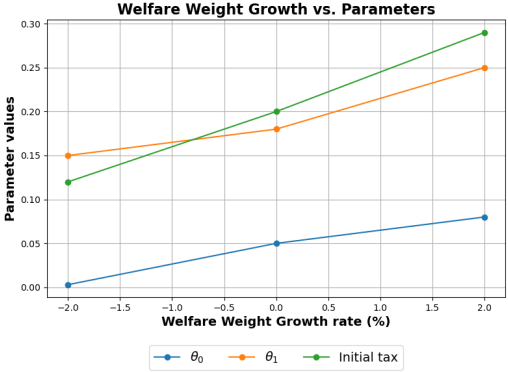
Mitigations

- Explicit benchmark tests and stress scenarios
- Mixed exogenous/endogenous sampling
- Report residual distributions and constraints
- Active learning and uncertainty-aware evaluation

Script Tbl. 12.3 lists the complementary *training-time* failure modes (loss declines but solution wrong; policy collapse; NaN; large boundary Euler errors; GP variance too large).

Representative Constraint and Uncertainty Results





Left: SCC uncertainty decomposition (Friedl et al., 2023). Right: constrained policy frontier (Kübler, Scheidegger & Surbek, 2026 forthcoming, *J. Political Economy: Macroeconomics*; arXiv:2507.01704).

Implementation Checklist

1. Define state/control vectors and feasibility transforms.
2. Engineer residual blocks from economics and climate equations.
3. Set sampling scheme and training diagnostics before tuning architecture.
4. Validate on out-of-sample simulations and benchmark moments.
5. Build surrogate/UQ layer only after solver quality is verified.
6. Separate normative objective from political/participation constraints.

Research Frontier

- Richer heterogeneity: regional, sectoral, and household dimensions.
- Endogenous innovation and technology diffusion under climate policy.
- Multi-model and model-ensemble learning for robust policy design.
- Tight integration of Bayesian learning and policy commitment under ambiguity.
- Better theoretical guarantees for convergence and error bounds.

Reading and Notebook Map

Climate readings

- `readings/links_by_lecture/lecture_16.md` (CDICE; Folini et al. 2025, *Restud*)
- `readings/links_by_lecture/lecture_17.md` (Deep UQ + Pareto carbon tax; KSS 2026 forthcoming, JPE: Macro)

Climate notebooks (suggested order)

- `01_Climate_Exercise.ipynb` (NumPy warm-up: carbon cycle, damages, BAU vs mitigation)
- `02_DICE_DEQN_Library_Port.ipynb` (deterministic CDICE-DEQN, library port)
- `03_Stochastic_DICE_DEQN.ipynb` (AR(1) shocks, Gauss–Hermite, SCC fan charts)

Closing Message

From method to policy

Deep learning methods are most useful when they improve economic credibility, computational scalability, and policy transparency at the same time.

Thank you for participating in the course.

Questions and discussion